

Cybersecurity Training Diary

Day 6: Firewalls & IDS/IPS

1. Topics Covered

✓ Firewalls Explained

Type	Function	Deployment	Pros	Cons
Packet Filtering	Checks headers (IP/port)	Network edge	Fast	No app awareness
Stateful	Tracks connections	Perimeter	Context-aware	Resource-heavy
Next-Gen (NGFW)	Deep packet inspection	Cloud/on-prem	App-layer security	Expensive
WAF	Web app protection	App frontend	SQLi/XSS protection	False positives

✓ IDS vs IPS

Feature	IDS (Detection)	IPS (Prevention)
Action	Alerts only	Blocks threats
Placement	Offline (TAP)	Inline
Speed Impact	None	Minimal latency
Best For	Monitoring	Active protection

✓ Real-World Tools

- **Firewalls:** pfSense, Cisco ASA, AWS Security Groups
- **IDS/IPS:** Snort, Suricata, Palo Alto Threat Prevention

2. Key Takeaways

Layered Defense: Use both firewall + IDS/IPS

NGFWs understand applications, not just ports

IPS false positives can break legitimate traffic

3. Practical Exercises

Exercise 1: Configure pfSense Firewall Rules

1. Block all inbound traffic except HTTP/HTTPS
2. Create an alert rule for port scans

Exercise 2: Snort IDS Rule Creation

```
alert tcp any any -> $HOME_NET 22 (msg:"SSH Bruteforce"; threshold:type t
```

4. Questions for Reflection

When would you choose an IDS over an IPS?

How could an attacker bypass a stateful firewall?

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